

Theory questions for practice

1. With a neat, labelled diagram of pn junction diode at equilibrium, derive an expression for barrier potential.
2. Explain working of pn junction diode in forward and reverse bias.
3. Plot diode characteristics for forward and reverse bias. What are static and dynamic resistance. Write their formulae.
4. Explain working of pn junction as half wave rectifier.
5. Explain working of pn junction as full wave rectifier.
6. Write note of Zener diode. How it is used as regulator
7. What is Schottky diode? Explain its working and state two applications.
8. Explain in brief the construction and working of PNP or NPN transistors.
9. What are the various configurations in which transistors are used? Draw appropriate diagrams.
10. What are the current gains for Common Base (CB) and Common Emitter (CE) configurations of transistor? Obtain relations between them.
11. Draw circuit diagram and explain the typical input and output characteristics of CB configuration of Bipolar Junction Transistor (BJT). Mention few salient features.
12. Draw circuit diagram and explain the typical input and output characteristics of CE configuration of BJT
13. What is the early effect? What is Q point? Explain its significance briefly. Draw necessary diagram?
14. With appropriate diagrams, explain working of BJT as the switch.
15. What is Metal Oxide Semiconductor (MOS)? Explain its working and state few applications
16. What is Charge Storage Mechanism in Metal Oxide Semiconductor Flash Memory Device
17. Explain construction and working of Light Emitting Diode. state its three advantages and disadvantages.
18. Explain construction and working of a laser diode. Give applications of Laser Diode

19. Explain working principal of photodetector. What are different types of Photodetectors
20. With neat diagram explain construction and working of a solar cell
21. Write five points to distinguish between laser diode and LED.